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FRANCESCO GRUSSU

CONTACT AND ONLINE PROFILES

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EDUCATION AND TRAINING

2012-2016 — University College London (UCL), UK

PhD in MR Physics

Mark: viva passed with no corrections. Award date: 28/03/2016.
Thesis: "*Microstructural imaging of the human spinal cord with advanced diffusion MRI*"

2010-2012 — University of Genoa, Italy

Master's Degree in Bioengineering

Mark: 110 out of 110 *cum laude* & *Dignità di Stampa* (Examination Panel award).
Dissertation: "*A study on a bidirectional brain-machine interface inspired by the corticospinal control of movement*".

2006-2009 — University of Cagliari, Italy

Bachelor's Degree in Biomedical Engineering

Mark: 110 out of 110 *cum laude*.
Dissertation: "*Real time wavelet denoising on a DSP of neural signals coming from the peripheral nervous system*".

WORK EXPERIENCE

Jan.2025-now — Senior investigator, VHIO — Spain

Vall d'Hebron Institute of Oncology

Affiliations: Radiomics Group, Biomedical Research Department.
Role: supervisor of research line in diffusion MRI. Design of MRI studies.

Oct.2020-Dec.24 — Senior post-doc, VHIO — Barcelona, Spain

Vall d'Hebron Institute of Oncology

Affiliations: Radiomics Group, Clinical Research Department.
Role: development of microstructural MRI techniques in cancer.

Feb.2016-Sept.20 — Research Associate, UCL — London, UK

University College London

Affiliations: Institute of Neurology; Centre for Medical Image Computing.
Responsibilities: quantitative MRI development (spinal cord, brain, prostate).

May-June 2012 — Research Assistant, QMUL — London, UK

Queen Mary Univ. of London

Affiliation: School of Electronic Engineering and Computer Science (EECS).
Responsibilities: image segmentation (Computed Tomography Angiography).

RESEARCH VISITS

Feb.-March 2024 — Champalimaud Foundation, Portugal

Visiting Researcher

Affiliation: Preclinical MRI lab, Champalimaud Foundation, Lisbon, Portugal.
Training: preclinical MRI and multi-echo gradient echo imaging.

Sept.-Nov. 2017 — New York University (NYU), USA

Visiting Researcher

Affiliation: Radiology, Langone Medical Center, New York City (USA).
Training: advanced denoising techniques for diffusion MRI.

IMPACT STATS ON THE 07/11/2026

h-index

Google Scholar h-index: 27. Web of Science h-index: 21.

Citations

Total Google Scholar citations: 3082. Web of Science citations: 1822.

FUNDING: PROJECT GRANTS

EU4Health 2026

EU4Health project grant, call EU4H-2026-SANTE-PJ-08, European Commission. "*Scale-up biotech AI innovation by leveraging responsible data use under the European Health Data Space*" (SCAILED). **Award**: 92,362.4 €. **Role**: VHIO official co-investigator. Principal investigator and primary coordinator: Gu Wei, Luxembourg National Data Service (LNDS). **Consortium**: 47 institutes/centres, with total budget of 14,384,759.99 €. **Duration**: 01/10/2026-30/09/2029.

CaixaResearch 2025

CaixaResearch Institute Innovation Hub 2025. CaixaResearch Institute, Barcelona (Spain). "*Computational Magnetic Resonance for metastatic prostate cancer*". **Award**: 30,000 €. **Role**: principal investigator. **Duration**: 01/01/2026-31/12/2026. Competitive call open to researchers from Innovation Hub institutions (VHIO; IrsiCaixa; ISGlobal; BarcelonaBeta).

AEI Generación de Conocimiento/ Knowledge Generation 2024

2024 Knowledge Generation Projects (*Proyectos de Generación de Conocimiento*), National Research Agency (*Agencia Estatal de Investigación* (AEI)), Ministry of Science, Innovation and Universities, Spain. "*Magnetic Resonance Imaging (MRI) foundational artificial intelligence for non-invasive, histologically-meaningful cancer characterisation (MRI-Found-Histo)*". **Award**: 118,750 €. **Role**: principal investigator. **Duration**: 01/09/2025-31/08/2028. **Code**: PID2024-158670OA-I00.

UCL pump-priming award 2017

Departmental seed grant. Centre for Medical Image Computing *Pump-priming Award* at University College London (UCL). "*Enabling multi-site high precision spinal cord MRI*". **Award**: GBP 23,900. **Role**: principal investigator. **Duration**: 01/07/2017-30/06/2018.

FUNDING: FELLOWSHIPS AND STUDENTSHIPS

AEI "Ramón y Cajal" Fellowship 2025

2025 Ramón y Cajal tenure-track fellowship, National Research Agency (*Agencia Estatal de Investigación* (AEI)), Ministry of Science, Innovation and Universities, Spain. "*Computational Magnetic Resonance biophysics for enhanced neurological and oncological imaging*". **Award**: €259,810. **Role**: principal investigator and fellow. **Duration**: 2027-2031 (awarded on 07/09/2026, agreement with Universitat Politècnica de Catalunya submitted to AEI on 28/09/2026). **Code**: RYC2025-054665-I.

"la Caixa" Junior Leader Fellowship 2022

2022 Junior Leader Retaining post-doctoral fellowship, "la Caixa" Foundation, Spain. "*New-generation oncological MRI (New-OncoMRI): development, validation and application*". **Award**: €294,900. **Role**: principal investigator and fellow. **Duration**: 30/09/2022-29/09/2025. **Code**: ID 100010434, fellowship number LCF/BQ/PR22/11920010.

AGAUR Beatriu de Pinós Fellowship 2020

2020 Beatriu de Pinós post-doctoral fellowship, AGAUR, Secretary of Universities and Research (Govt of Catalonia, Spain). "*Advancing Magnetic Resonance Imaging against liver cancer*". **Award**: €144,300. **Role**: principal investigator and fellow. **Duration**: 01/01/2022-31/12/2024, renounced on 29/09/2022 due to incompatibility with the "la Caixa" Junior Leader fellowship. **Code**: 2020 BP 00117.

UCL Grand Challenge PhD studentship 2012

UCL School of Life and Medical Science *Grand Challenge PhD Studentship*, awarded to work on the project "*Axonal density as MR imaging biomarker: from bench to bedside*". **Award:** 59,000 GBP. **Description:** PhD studentship covering a 3-year student stipend, tuition fees at UK/EU rate, and research costs. **Duration:** 09/2012-12/2015.

POSITIONS: DECLINED COMPETITIVE AWARDS

2025 UAB "Serra Húnter" tenure-track Lecturership

Serra Húnter tenure-track Lecturership/Assistant Professorship (*professor ajudant doctor*) funded by the Generalitat de Catalunya. Department of Computer Sciences, Universitat Autònoma de Barcelona, Cerdanyola/Barcelona, Spain. **Duration**: 2026-2032 (awarded 30/06/2026, declined due to incompatibility with RyC fellowship). **Code**: UAB-LE-251-010.

SCIENTIFIC ARTICLES AND REVIEWS — SENIOR AUTHORSHIP

Magn Reson Med 2026

"*Simulation-informed evaluation of microvascular parameter mapping for diffusion MR imaging of solid tumours*". Voronova AK et al, Grussu F. Magnetic Resonance in Medicine (2026), 96(1): 387-402, doi: 10.1002/mrm.70318. Senior (last) author, corresponding author. **IF**: 3.3 (Quartile Q2) in the year of publication.

Comm Bio 2025

"*Histology-informed microstructural diffusion simulations for MRI cancer characterisation — the Histo-microSim framework*". Grigoriou A et al, Grussu F. Communications Biology (2025), 8: 1695, doi: 10.1038/s42003-025-09096-3. Senior (last) author, corresponding author. **IF**: 5.1 (Decile D1) in the year of publication.

MedIA 2025

"*SpinFlowSim: a blood flow simulation framework for histology-informed diffusion MRI microvasculature mapping in cancer*". Voronova AK et al, Grussu F. Medical Image Analysis (2025), 102: 103531, doi: 10.1016/j.media.2025.103531. Senior (last) author, corresponding author. **IF**: 11.8 (Decile D1) in the year of publication.

JMRI 2025

"*Enhancing tumor microstructural quantification with machine learning and diffusion-relaxation MRI* ". Macarro C et al, Grussu F. Journal of Magnetic Resonance Imaging (2025), 61(2): 1018-1021, doi: 10.1002/jmri.29484. Senior (last) author, corresponding author. **IF**: 3.5 (Quartile Q1) in the year of publication.

JMRI 2024

Review: "*Advanced diffusion-weighted MRI for cancer microstructure assessment in body imaging, and its relationship with histology*". Fokkinga E, Hernandez-Tamames JA, Ianus A, Nilsson M, Tax CMW, Perez-Lopez R, Grussu F. Journal of Magnetic Resonance Imaging (2024), 60(4): 1278-1304, doi: 10.1002/jmri.29144. Senior (last) author, corresponding author. **IF**: 3.3 (Quartile Q1) in the year of publication.

Front Neurol 2021

"*Comparison of neurite orientation dispersion and density imaging and two-compartment spherical mean technique parameter maps in multiple sclerosis*". Johnson D et al, Grussu F. Frontiers in Neurology (2021), 12: 662855, doi: 10.3389/fneur.2021.662855. Senior (last) author. **IF**: 4.086 (Quartile Q2) in the year of publication.

SCIENTIFIC ARTICLES AND REVIEWS — (CO)-FIRST AUTHORSHIP

J Immunother Cancer 2025

"*Integrating C-reactive protein flare and early MRI dynamics for enhanced prediction of*

immunotherapy response". Navarro D*, Grussu F* et al. Journal for ImmunoTherapy of Cancer (2025), 13: e012143, doi: 10.1136/jitc-2025-012143. *: DN and FG are joint first authors (equal contribution). **IF**: 10.6 (Decile D1) in the year of publication.

Comm Med 2025

"*Clinically feasible liver tumour cell size measurement through histology-informed in vivo diffusion MRI*". Grussu F et al. Communications Medicine (2025), 5: 535, doi: 10.1038/s43856-025-01246-2. Corresponding author. **IF**: 6.3 (Quartile Q1) in the year of publication.

Cell Rep Med 2024

"*An accessible deep learning tool for voxel-wise classification of brain malignancies from perfusion MRI*". Garcia-Ruiz A*, Pons-Escoda A*, Grussu F* et al. Cell Reports Medicine (2024), 5(3): 101464, doi: 10.1016/j.xcrm.2024.101464. *: AGR, APE and FG are joint first authors (equal contribution). **IF**: 11.7 (Decile D1) in the year of publication.

Magn Reson Med 2022

"*Diffusion MRI signal cumulants and hepatocyte microstructure at fixed diffusion time: Insights from simulations, 9.4T imaging, and histology*". Grussu F et al. Magnetic Resonance in Medicine (2022), 88(1): 365-379, doi: 10.1002/mrm.29174. Corresponding author. **IF**: 3.737 (Quartile Q2) in the year of publication.

Front Phys 2021

"*Feasibility of data-driven, model-free quantitative MRI protocol design: application to brain and prostate diffusion-relaxation imaging*". Grussu F et al. Frontiers in Physics (2021), 9: 752208, doi: 10.3389/fphy.2021.752208. Corresponding author. **IF**: 3.560 (Quartile Q2) in the year of publication.

Sem Ultrasound CT MRI 2021

Review: "*Diffusion-weighted imaging: recent advances and applications*". Martinez-Heras E*, Grussu F*, et al. Seminars in Ultrasound, CT and MRI (2021), 42(5): 490-506, doi: 10.1053/j.sult.2021.07.006. *: EMH and FG are joint first authors (equal contribution). **IF**: 1.600 (Quartile Q3) in the year of publication.

NeuroImage 2020

"*Multi-parametric quantitative in vivo spinal cord MRI with unified signal readout and image denoising*". Grussu F et al. NeuroImage (2020), 217: 116884, doi: 10.1016/j.neuroimage.2020.116884. Corresponding author. **IF**: 5.902 (Decile D1) in the year of publication.

Magn Reson Med 2019

"*Relevance of time-dependence for clinically viable diffusion imaging of the spinal cord*". Grussu F et al. Magnetic Resonance in Medicine (2019), 81(2): 1247-1264, doi: 10.1002/mrm.27463. Corresponding author. **IF**: 3.858 (Quartile Q1) in the year of publication.

Ann Clin Transl Neurol 2017

"*Neurite dispersion: a new marker of multiple sclerosis spinal cord pathology?*". Grussu F, Schneider T et al. Annals of Clinical and Translational Neurology (2017), 4(9):663-679, doi: 10.1002/acn3.445. FG and TS are joint first authors (equal contribution). **IF**: 3.901 (Quartile Q1) in the year of publication. Paper featured in Nature Reviews Neurology "Research Highlights" (Patel M, Nat Rev Neur (2017), 13(10): 578, doi: 10.1038/nrneurol.2017.127).

J Neurosci Meth 2016

"*A framework for optimal whole-sample histological quantification of neurite orientation dispersion in the human spinal cord*". Grussu F et al. Journal of Neuroscience Methods (2016), 273: 20-32, doi: 10.1016/j.jneumeth.2016.08.002. Corresponding author. **IF**: 2.053 (Quartile Q3) in the year of publication.

NeuroImage 2015

"*Neurite orientation dispersion and density imaging of the healthy cervical spinal cord* in vivo". Grussu F et al. NeuroImage (2015), 111: 590-601, doi: 10.1016/j.neuroimage.2015.01.045. Corresponding author. **IF**: 6.357 (Decile D1) in the year of publication.

SCIENTIFIC ARTICLES AND REVIEWS — CO-AUTHORSHIP

Eur Radiol Exp 2026

"*Validating the biological origin of in vivo fractional intracellular volume from VERDICT-MRI in the prostate*". Masramon M, ..., Grussu F et al. European

Radiology Experimental (2026), 10: 23, doi: 10.1186/s41747-026-00797-w. **IF**: 4.7 (Quartile Q1) in the year of publication.

Eur Radiol 2026

Review: "*ESR Essentials: diffusion-weighted MRI—practice recommendations by the European Society for Magnetic Resonance in Medicine and Biology*". Palombo M, Bodini B, Grussu F et al. European Radiology (2026), 36: 2198-2208, doi: 10.1007/s00330-025-12033-x. **IF**: 6.0 (Quartile Q1) in the year of publication.

JMRI 2026

"*The sense of smell (SoS) atlas: its creation and first application to investigate COVID-19 related anosmia with a comprehensive quantitative MRI protocol*". Gaviraghi M, ..., Grussu F et al. Journal of Magnetic Resonance Imaging (2026), 63(2): 574-593, doi: 10.1002/jmri.70128. **IF**: 3.5 (Quartile Q1) in the year of publication.

Clin Cancer Res 2026

"*Broad utility of ultrasensitive analysis of ctDNA dynamics across solid tumors treated with immunotherapy*". Garralda E, ..., Grussu F et al. Clinical Cancer Research (2026), 32(2): 333-349, doi: 10.1158/1078-0432.CCR-25-2312. **IF**: 10.2 (Decile D1) in the year of publication.

Magn Res Med 2025c

Review: "*Considerations and recommendations from the ISMRM Diffusion Study Group for preclinical diffusion MRI: Part 3—Ex vivo imaging: Data processing, comparisons with microscopy, and tractography*". Schilling KG, Howard AFD, Grussu F et al. Magnetic Resonance in Medicine (2025), 93(6): 2561-2582, doi: 10.1002/mrm.30424. **IF**: 3.0 (Quartile Q2) in the year of publication.

Magn Res Med 2025b

Review: "*Considerations and recommendations from the ISMRM diffusion study group for preclinical diffusion MRI: Part 2—Ex vivo imaging: Added value and acquisition*". Schilling KG, Grussu F et al. Magnetic Resonance in Medicine (2025), 93(6): 2535-2560, doi: 10.1002/mrm.30435. **IF**: 3.0 (Quartile Q2) in the year of publication.

Magn Res Med 2025a

Review: "*Considerations and recommendations from the ISMRM diffusion study group for preclinical diffusion MRI: Part 1: In vivo small-animal imaging*". Jelescu IO, Grussu F et al. Magnetic Resonance in Medicine (2025), 93(6): 2507-2534, doi: 10.1002/mrm.30429. **IF**: 3.0 (Quartile Q2) in the year of publication.

Imag Neurosci 2025

"*Body size and intracranial volume interact with the structure of the central nervous system: A multi-center in vivo neuroimaging study*". Labounek R, ... Grussu F et al. Imaging Neuroscience (2025), 3: 00559, doi: 10.1162/imag_a_00559. No IF available in the year of publication.

Sci Rep 2025

"*Evaluation of magnetic resonance spectroscopy total sodium concentration measures, and associations with microstructure and physical impairment in cervical myelopathy*". Solanky B, ... Grussu F et al. Scientific Reports (2025), 15: 7014, doi: 10.1038/s41598-025-91658-w. **IF**: 3.9 (Quartile Q1) in the year of publication.

J Immunother Cancer 2025

"*Radiomics signature for dynamic monitoring of tumor inflamed microenvironment and immunotherapy response prediction*". Bernatowicz K, ... Grussu F et al. Journal for ImmunoTherapy of Cancer (2025), 13: e009140, doi: 10.1136/jitc-2024-009140. **IF**: 10.6 (decile D1) in the year of publication.

NeuroImage Rep 2024

"*Investigating the relationship between thalamic iron concentration and disease severity in secondary progressive multiple sclerosis using quantitative susceptibility mapping: Cross-sectional analysis from the MS-STAT2 randomised controlled trial*". Williams T, ..., Grussu F, ..., Chataway J; On behalf of The UCL MS-STAT2 investigators. NeuroImage: Reports (2024), 4(3): 100216, doi: 10.1016/j.ynirp.2024.100216. No IF available in the year of publication.

Eur Urol 2024

"*Whole-body magnetic resonance imaging as a treatment response biomarker in castration-resistant prostate cancer with bone metastases: the iPROMET clinical trial*". Garcia-Ruiz A, ... Grussu F et al. European Urology (2024), 86(3): 272-274, doi: 10.1016/j.eururo.2024.02.016. **IF**: 25.3 (Decile D1) in the year of publication.

Mult Scler 2024

"*What contributes to disability in progressive MS? A brain and cervical cord–matched quantitative MRI study*". Tur C, ..., Grussu F et al. Multiple Sclerosis Journal

(2024), e-pub ahead of print, doi: 10.1177/13524585241229969. **IF**: 4.8 (Quartile Q1) in the year of publication.

Radiology: AI 2024 — "*Identification of precise 3D CT radiomics for habitat computation by machine learning in cancer*". Prior O, ..., Grussu F et al. Radiology: Artificial Intelligence (2024), 6(2): e230118, doi: 10.1148/ryai.230118. **IF**: 8.1 (Quartile Q1) in the year of publication.

Movement Disord 2023 — "*Multimodal analysis of the visual pathways in Friedreich's Ataxia reveals novel biomarkers*". Thomas-Black G, ... Grussu F et al. Movement Disorders (2023), 38(6): 959-969, doi: 10.1002/mds.29277. **IF**: 8.6 (Decile D1) in the year of publication.

Sci Rep 2023 — "*Feasibility of in vivo multi-parametric quantitative magnetic resonance imaging of the healthy sciatic nerve with a unified signal readout protocol*". Boonsuth R, Battiston M, Grussu F et al. Scientific Reports (2023), 13: 6565, doi: 10.1038/s41598-023-33618-w. **IF**: 4.6 (Quartile Q2) in the year of publication.

Front Neuroinform 2023 — "*Patterns of inflammation, microstructural alterations, and sodium accumulation define multiple sclerosis subtypes after 15 years from onset*". Ricciardi A, Grussu F et al. Frontiers in Neuroinformatics (2023), 17: 1060511, doi: 10.3389/fninf.2023.1060511. **IF**: 3.5 (Quartile Q2) in the year of publication.

Neurology 2023 — "*Differentiating Multiple Sclerosis from AQP4-Neuromyelitis Optica Spectrum Disorder and MOG-antibody disease with imaging*". Cortese R, ... Grussu F et al. Neurology (2023), 100(3): e308-e323, doi: 10.1212/WNL.0000000000201465. **IF**: 10.1 (Decile D1) in the year of publication.

Am J Neuroradiol 2022 — "*Diffuse large B-cell Epstein-Barr virus–positive primary CNS lymphoma in non-AIDS patients: high diagnostic accuracy of DSC perfusion metrics*". Pons-Escoda A, ... Grussu F et al. American Journal of Neuroradiology (2022), 43(11): 1567-1574, doi: 10.3174/ajnr.A7668. **IF**: 4.966 (Quartile Q2) in the year of publication.

Magn Reson Med 2022c — "*Multi-echo quantitative susceptibility mapping: how to combine echoes for accuracy and precision at 3 Tesla*". Biondetti E, ..., Grussu F et al. Magnetic Resonance in Medicine (2022), 88(5): 2101-2116, doi: 10.1002/mrm.29365. **IF**: 3.737 (Quartile Q2) in the year of publication.

Magn Reson Med 2022b — "*SENSE EPI reconstruction with 2D phase error correction and channel-wise noise removal*". Powell E, ..., Grussu F et al. Magnetic Resonance in Medicine (2022), 88(5): 2157-2166, doi: 10.1002/mrm.29349. **IF**: 3.737 (Quartile Q2) in the year of publication.

Magn Reson Med 2022a — "*Comparison of multicenter MRI protocols for visualizing the spinal cord gray matter*". Cohen-Adad J, ..., Grussu F et al. Magnetic Resonance in Medicine (2022), 88(2): 849-859, doi: 10.1002/mrm.29249. **IF**: 3.737 (Quartile Q2) in the year of publication.

Eur Radiol 2022 — "*Voxel-level analysis of normalized DSC-PWI time-intensity curves: a potential generalizable approach and its proof of concept in discriminating glioblastoma and metastasis*". Pons-Escoda A, ..., Grussu F et al. European Radiology (2022), 32: 3705–3715, doi: 10.1007/s00330-021-08498-1. **IF**: 7.034 (Quartile Q1) in the year of publication.

BMJ Open 2022 — "*Histo-MRI map study protocol: a prospective cohort study mapping MRI to histology for biomarker validation and prediction of prostate cancer*". Singh S, ..., Grussu F et al. BMJ Open (2022), 12: e059847, doi: 10.1136/bmjopen-2021-059847. **IF**: 3.007 (Quartile Q2) in the year of publication.

NeuroImage Clin 2022 — "*Spatial patterns of brain lesions assessed through covariance estimations of lesional voxels in multiple sclerosis: the SPACE-MS technique*". Tur C, Grussu F et al. NeuroImage: Clinical (2022), 33: 102904, doi: 10.1016/j.nicl.2021.102904. **IF**: 4.891 (Quartile Q2) in the year of publication.

Front Neurol 2021 — "*Assessing lumbar plexus and sciatic nerve damage in relapsing-remitting multiple*

sclerosis using magnetisation transfer ratio". Boonsuth R, ..., Grussu F et al. Frontiers in Neurology (2021), 12: 763143, doi: 10.3389/fneur.2021.763143. **IF**: 4.003 (Quartile Q2) in the year of publication.

Sci Rep 2021 — "*Robust imaging habitat computation using voxel-wise radiomics features*". Bernatowicz K, Grussu F et al. Scientific Reports (2021), 11: 20133, doi: 10.1038/s41598-021-99701-2. **IF**: 4.380 (Quartile Q1) in the year of publication.

Sci Data 2021 — "*Open-access quantitative MRI data of the spinal cord and reproducibility across participants, sites and manufacturers*". Cohen-Adad J, ..., Grussu F et al. Scientific Data (2021), 8: 219, doi: 10.1038/s41597-021-00941-8. **IF**: 6.444 (Quartile Q1) in the year of publication.

Nat Protoc 2021 — "*Generic acquisition protocol for quantitative MRI of the spinal cord*". Cohen-Adad J, ..., Grussu F et al. Nature Protocols (2021), 16: 4611–4632, doi: 10.1038/s41596-021-00588-0. **IF**: 13.491 (decile D1) in the year of publication.

Brain 2021 — "*Brain microstructural and metabolic alterations detected in vivo at the onset of the first demyelinating event*". Collorone S, ..., Grussu F et al. Brain (2021), 144: 1409-1421, doi: 10.1093/brain/awab043. **IF**: 13.501 (Decile D1) in the year of publication.

NeuroImage 2021 — "*Uncertainty modelling in deep learning for safer neuroimage enhancement: demonstration in diffusion MRI*". Tanno R, ... Grussu F et al. NeuroImage (2021), 225: 117366, doi: 10.1016/j.neuroimage.2020.117366. **IF**: 6.556 (Decile D1) in the year of publication.

Mult Scler 2020b — "*Reduced neurite density in the brain and cervical spinal cord in relapsing–remitting multiple sclerosis: A NODDI study*". Collorone S, Cowley N, Grussu F et al. Multiple Sclerosis Journal (2020), 26(13): 1647-1657, doi: 10.1177/1352458519885107. **IF**: 5.412 (Quartile Q1) in the year of publication.

Mult Scler 2020a — "*A multi-shell multi-tissue diffusion study of brain connectivity in early multiple sclerosis*". Tur C, Grussu F et al. Multiple Sclerosis Journal (2020), 26(7): 774-785, doi: 10.1177/1352458519845105. **IF**: 5.412 (Quartile Q1) in the year of publication.

NeuroImage 2020b — "*Cross-scanner and cross-protocol multi-shell diffusion MRI data harmonization: Algorithms and results*". Ning L, Bonet-Carne E, Grussu F et al. NeuroImage (2020), 221: 117128, doi: 10.1016/j.neuroimage.2020.117128. **IF**: 5.902 (Decile D1) in the year of publication.

NeuroImage 2020a — "*Generalised boundary shift integral for longitudinal assessment of spinal cord atrophy*". Prados F, ..., Grussu F et al. NeuroImage (2020), 209: 116489, doi: 10.1016/j.neuroimage.2019.116489. **IF**: 5.902 (Decile D1) in the year of publication.

Magn Reson Med 2019 — "*Fast bound pool fraction mapping via steady-state magnetization transfer saturation using single-shot EPI*". Battiston M, ..., Grussu F et al. Magnetic Resonance in Medicine (2019), 82: 1025-1040, doi: 10.1002/mrm.27792. **IF**: 3.858 (Quartile Q1) in the year of publication.

NeuroImage 2019 — "*Cross-scanner and cross-protocol diffusion MRI data harmonisation: a benchmark database and evaluation of algorithms*". Tax CMW, Grussu F et al. NeuroImage (2019), 195: 285-299, doi: 10.1016/j.neuroimage.2019.01.077. **IF**: 5.812 (Decile D1) in the year of publication.

Sci Rep 2018 — "*Structural cortical network reorganization associated with early conversion to multiple sclerosis*". Tur C, ..., Grussu F et al. Scientific Reports (2018), 8: 10715, doi: 10.1038/s41598-018-29017-1. **IF**: 4.122 (Quartile Q1) in the year of publication.

Magn Reson Med 2018b — "*An optimized framework for quantitative magnetization transfer imaging of the cervical spinal cord in vivo*". Battiston M, Grussu F et al. Magnetic Resonance in Medicine (2018) 79(5): 2576-2588, doi: 10.1002/mrm.26909. **IF**: 3.924 (Quartile Q1) in the year of publication.

Magn Reson Med 2018a

"*Fast and reproducible in vivo T1 mapping of the human cervical spinal cord*". Battiston M, ..., Grussu F et al. Magnetic Resonance in Medicine (2018), 79(4): 2142-2148, doi: 10.1002/mrm.26852. **IF**: 3.924 (Quartile Q1) in the year of publication.

NeuroImage 2017

"*Spinal cord grey matter segmentation challenge*". Prados F, ..., Grussu F et al. NeuroImage (2017), 152:312–329, doi: 10.1016/j.neuroimage.2017.03.010. **IF**: 5.835 (Decile D1) in the year of publication.

PlosOne 2016

"*Reduced field-of-view diffusion-weighted imaging of the lumbosacral enlargement: a pilot in vivo study of the healthy spinal cord at 3T*". Yiannakas MC, Grussu F et al. PlosOne (2016), 11(10): e0164890, doi: 10.1371/journal.pone.0164890. **IF**: 3.057 (Quartile Q1) in the year of publication.

CONFERENCE PAPERS: FIRST AUTHORSHIP

CDMRI 2020

"*Deep learning model fitting for diffusion-relaxometry: a comparative study*". Grussu F, Battiston M, Palombo M, Schneider T, Wheeler-Kingshott CAM, Alexander DC. Proc of 2020 MICCAI Workshop on Computational Diffusion MRI, 2021, 159-172, doi: 10.1007/978-3-030-73018-5_13. Corresponding author.

CONFERENCE PAPERS: CO-AUTHORSHIP

MICCAI 2022

"*Progressive subsampling for oversampled data - application to quantitative MRI*". Blumberg SB, ..., Grussu F et al. Proc of Medical Image Computing and Computing Assisted Intervention (MICCAI) 2022, Lecture Notes in Computer Science, 13436: 421–431, doi: 10.1007/978-3-031-16446-0_40.

CDMRI 2019

"*Acquiring and predicting multidimensional diffusion (MUDI) data: an open challenge*". Pizzolato M, ..., Grussu F et al. Proc of 2019 MICCAI Workshop on Computational Diffusion MRI, 2020, 195-208, doi: 10.1007/978-3-030-52893-5_17.

CDMRI 2018b

"*Multi-shell diffusion MRI harmonisation and enhancement challenge (MUSHAC): progress and results*". Ning L, ..., Grussu F et al. Proc of 2018 MICCAI Workshop on Computational Diffusion MRI, 2019, 217-224, doi: 10.1007/978-3-030-05831-9_18.

CDMRI 2018a

"*Spatial characterisation of fibre response functions for spherical deconvolution in multiple sclerosis*". Tur C, Grussu F et al. Proc of 2018 MICCAI Workshop on Computational Diffusion MRI, 2019, 265-279, doi: 10.1007/978-3-030-05831-9_21.

MICCAI 2016

"*Bayesian image quality transfer*". Tanno R, ..., Grussu F et al. Proc of Medical Image Computing and Computing Assisted Intervention (MICCAI) 2016, Lecture Notes in Computer Science, 9901: 265-273, doi: 10.1007/978-3-319-46723-8_31.

BOOK CHAPTERS

CRC Press 2018

Chapter 8: "*D — the diffusion of water (DTI)*". Grussu F and Wheeler-Kingshott CAM. "*Quantitative MRI of the brain*" (2nd edition, 2018), Cercignani M, Dowell N and Tofts P editors. ISBN 978-1-138-03285-9, doi: 10.1201/b21837.

BOOK EDITING

CDMRI 2018

"*Computational Diffusion MRI*". Bonet-Carne E, Grussu F, Ning L, Sepehrband F and Tax C editors. Proc. of 2018 MICCAI Workshop on "*Computational Diffusion MRI*", Granada, Spain, 20/09/2018. ISBN: 978-3-030-05830-2, doi: 10.1007/978-3-030-05831-9.

CDMRI 2017 — "*Computational Diffusion MRI*". Kaden E, Grussu F, Ning L, Tax C and Veraart J editors. Proc. of 2017 MICCAI Workshop on "*Computational Diffusion MRI*", Quebec City, Canada, 10/09/2017. ISBN: 978-3-319-73839-0, doi: 10.1007/978-3-319-73839-0.

PRIZES AND AWARDS

2023 — **Best oral paper award**, 2023 annual meeting of the Iberian Chapter of the International Society for Magnetic Resonance in Medicine (ISMRM), Trainee competition for abstract Grussu F et al, Proc of ISMRM Iberian Chapt. 2023.

2021 — **3rd prize**, 2021 ISMRM MR of Cancer Study Group, Trainee competition for abstract Grussu F et al, Proc of ISMRM 2021, p.0699.

2021-2015 — ***Magna cum Laude*** awards at the ISMRM annual meeting in 2021, 2020, 2018 and 2015.

2020 — **2nd prize** (shared), 2020 ISMRM British and Irish Chapter *"Mansfield Research Innovation Award"* for abstract Grussu F et al, Proc of ISMRM 2020.

2019 — **1st prize** in the "*Multi-dimensional Diffusion Imaging*" (MUDI) challenge at 2019 CDMRI MICCAI Workshop (Shenzhen, China, 17/10/2019) (**Team**: Grussu F, Blumberg SB, Ianus A, Mertzanidou T, Alexander DC; **Method**: SARDU-Net).

AWARDS IN SUPERVISORY ROLES

2024 — ***Best application (poster)*** by the Diffusion Study Group of the International Society for Magnetic Resonance in Medicine (ISMRM) to student Anna Voronova, for abstract (Voronova A et al, and Grussu F, p.0124, ISMRM 2024).

2024 — ***Summa cum Laude*** award to student Anna Voronova, for abstract (Voronova A et al, and Grussu F, p.0124, ISMRM 2024).

2024 — ***Magna cum Laude*** award to student Athanasios Grigoriou, for abstract (Grigoriou A et al, and Grussu F, p.0699, ISMRM 2024).

ELECTED INTERNATIONAL POSITIONS

ISMRM 2018-2020 — **Elected trainee (student) representative** for the *White Matter Study Group* of the International Society for Magnetic Resonance in Medicine (ISMRM).

INVITED ORAL COMMUNICATIONS

ISMRM Iberian Chapter 2026 — "*From histology to radiological maps: a journey through MRI signal modelling and beyond*". International Society for Magnetic Resonance in Medicine (ISMRM), 2026 annual meeting of the Iberian Chapter. "Keynote lecture". University of Coimbra, Coimbra (Portugal), 02/07/2026.

BSC 2025 — "*Histology-informed computer simulations enhance Magnetic Resonance Imaging (MRI) virtual biopsies in cancer*". "Severo Ochoa Research Seminar", Barcelona Supercomputing Center (BSC), Barcelona, Spain, 13/11/2025.

ODELIA Summer School 2025 — "*An overview of Magnetic Resonance Imaging (MRI): from basic principles to cutting-edge applications*". ODELIA consortium on open source swarm learning to empower medical AI. "2025 ODELIA Summer School", Vall d'Hebron Institute of Oncology (VHIO), Barcelona, Spain, 18/09/2025.

ISMRM Iberian Chapter 2025 — "*Quantitative imaging and AI Automation*". International Society for Magnetic Resonance in Medicine (ISMRM), 2025 annual meeting of the Iberian Chapter. "2025 ISMRM Iberian Chapter pre-conference Summer School", Institut de Bioenginyeria de Catalunya (IBEC), Barcelona (Spain), 02/07/2025.

ISMRM Workshop 2025 — "*Challenges in body diffusion and how to overcome them*". International Society for Magnetic Resonance in Medicine (ISMRM) Workshop on: "40 years of Diffusion: past, present and future perspectives", Kyoto University, Kyoto (Japan), 18/02/2025.

EPFL 2024 — "*Diffusion MR signal modelling for oncological body imaging*". Visitors Talk, Centre for Biomedical Imaging (CIBM), École Polytechnique Fédérale de Lausanne (EPFL), Lausanne (Switzerland), 09/10/2024.

ESMRMB workshop 2024 — "*Modelling body microstructure*". ESMRMB 2024 pre-congress workshop "Leaps in Microstructure Imaging: Exploring New Horizons", Barcelona (Spain), 02/10/2024.

DiPy workshop 2024 — "*Advanced body diffusion MRI for oncological applications*". 2024 DiPy workshop, online, 14/03/2024.

ESMRMB workshop 2023 — "*Histology-informed body diffusion MRI in oncological applications*". ESMRMB 2023 pre-congress workshop "Frontiers in preclinical MRI", Basel (Switzerland), 04/10/2023.

ESMRMB lectures 2023 — "*Diffusion MRI in the body*". ESMRMB Lectures on MR, "Introduction to diffusion-weighted MR imaging and spectroscopy", Cardiff (UK), 07/09/2023.

DSG ISMRM 2023 — "*Emerging models in oncology*". "Virtual biopsies by diffusion MRI: are we there yet in oncology?", Virtual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM) Diffusion Study Group (DSG). Online, 19/04/2023.

UCL 2022b — "*Data-driven, model-free, deep learning approach for quantitative MRI protocol design*". "Microstructure Imaging Meets Machine Learning" (MIML) workshop, University College London, London (UK), 13/05/2022.

ISMRM 2022 — "*Modeling diffusion in cancer and body*". Educational session: "Diffusion", 2022 annual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM), London (UK), 07/05/2022.

University of Verona 2019 — "*Diffusion MRI data harmonisation*". 2019 School on Brain Connectomics, University of Verona (Italy), 24/09/2019.

ISMRM Italian Chapter 2018 — "*Axonal dispersion from diffusion MRI: a new marker of microstructural damage*". Italian Association for Magnetic Res. in Medicine, Padua (Italy), 10/05/2018.

MODERATION OF SCIENTIFIC SESSIONS

ECR 2026 — **Moderator** of European Society of Radiology (ESR) Research Committee Session 10 "Real-world applications of imaging biomarkers: from hypothesis to clinical practice", 05/03/2026, 2026 European Conference of Radiology (ECR), organised by the ESR in Vienna (Austria).

ISMRM Iberian Chapter 2025 — **Moderator** of oral scientific session "Clinical session", 03/07/2025, 2025 annual meeting of the Iberian Chapter of the International Society for Magnetic Resonance in Medicine (ISMRM), Barcelona (Spain).

ISMRM 2022 — **Moderator** of oral "power pitch" scientific session "Motion correction", 10/05/2022, 2022 annual meeting of the International Society for Magnetic Resonance in Medicine (ISMRM), London (UK).

ISMRM 2021b — **Moderator** of oral scientific session "Microstructure: Modelling Gray & White Matter Diffusion", 19/05/2021, 2021 virtual annual meeting of the ISMRM.

ISMRM 2021a — **Facilitator** of poster session "Diffusion Applications: Brain & Spine", 18/05/2021, 2021 virtual annual meeting of the ISMRM.

PARTICIPATION IN EVALUATION PANELS

Grants — Reviewer, Finland 2026

Grant reviewer for the Research Council of Finland, 2026 Call for Clinical Medicine Research (online evaluation).

PhD thesis — Reviewer, UAB 2026

PhD thesis reviewer for the Universitat Autònoma de Barcelona (UAB), Cerdanyola/Barcelona (Spain). **Thesis title:** "*New immunoPET radiopharmaceuticals for immune cell assessment in ulcerative colitis*". **Evaluation report submitted on:** 04/08/2026 (online). **Role:** external expert reviewer (before thesis defense). **Degree:** PhD in Medicine.

MSc thesis — Examiner, UAB 2026

MSc defense examiner at the Universitat Autònoma de Barcelona (UAB), Cerdanyola/Barcelona (Spain). **Thesis title:** "*Metabolic profiling and classification of brain tumours via cNMF analysis of MRS data*". **Exam date:** 14/07/2026 (in person at UAB). **Role:** defense examiner. **Degree:** MSc in Bioinformatics.

Grants — Reviewer, Finland 2025

Grant reviewer for the Research Council of Finland, 2025 Call for Clinical Research (online evaluation).

PhD thesis — Examiner, EPFL 2024

PhD defense at the École Polytechnique Fédérale de Lausanne (EPFL), Lausanne (Switzerland). **Thesis title:** "*Lymph node microstructure imaging using diffusion MRI: From simulations to acquisitions*". **Exam date:** 08/10/2024 (in person in Lausanne). **Role:** external defense examiner. **Degree:** PhD. **Department:** Center for Biomedical Imaging (CIBM), EPFL. Thesis link here.

ORGANISATION OF SCIENTIFIC EVENTS

UB Seminar 2026b

Institute of Neuroscience Seminar at *Universitat de Barcelona* 2026: "Clinical applications of microstructure imaging in neurological disorders using free gradient waveforms ", 28/10/2026. **Speaker**: Markus Nilsson. **Organisers and moderators**: Roser Sala-Llonch, Francesco Grussu.

UB Seminar 2026a

Institute of Neuroscience Seminar at *Universitat de Barcelona* 2026: "Quantifying blood-brain barrier permeability using MRI ", 24/04/2026. **Speaker**: Elizabeth Powell. **Organisers and moderators**: Roser Sala-Llonch, Francesco Grussu.

ISMRM Iberian Chapter 2025

ISMRM Iberian Chapter: 2025 annual meeting of the Iberian Chapter of the International Society for Magnetic Resonance in Medicine (ISMRM) (Barcelona, scheduled for 3-4/07/2025). **Organisers:** Marco Rius I (IBEC), Ortega Machuca MA (IBEC), Candiota AP (UAB), Julià-Sapé M (UAB), Muñoz Moreno E (IDIBAPS), Lope-Piedrafita S (UAB), Barba Vert I (UVic), Benito M (UCM), Grussu F (VHIO), Gómez Cabeza D (IBEC).

MIS ISMRM 2021

Member Initiated Symposium at ISMRM 2020: "Looking Beyond Axons: Imaging the Immune System in White Matter", 19/05/2021. **Organisers**: Cohen-Adad J, Grussu F, Kolind S.

WMSG ISMRM 2019

ISMRM White Matter Study Group Virtual meeting: "Myelin Imaging in the Spinal Cord at High Field", 27/06/2019, joint meeting with the High-field Study Group. **Organiser**: Grussu F. **Chair**: Cohen-Adad J.

MIS ISMRM 2019

Member Initiated Symposium at ISMRM 2019: "Completing the Circle: Moving Multi-Parametric Neuro MRI into Clinical Practice and Trials", 15/05/2019. **Organisers**: Vrenken H, Cohen-Adad J, Grussu F.

CDMRI 2018

MICCAI Workshop: Computational Diffusion MRI (CDMRI) 2018 (Granada, Spain, 20/09/2018). **Organisers:** Bonet-Carne E (UCL), Grussu F (UCL), Ning L (Harvard), Sepehrband F (USC), Tax C (Cardiff University).

MUSHAC Challenge 2018

MICCAI Challenge: Multi-shell dMRI harmonisation and enhancement (MUSHAC, part of CDMRI 2018). **Organisers:** Bonet-Carne E (UCL), Grussu F (UCL), Ning L (Harvard), Sepehrband F (USC), Tax C (Cardiff University).

CDMRI 2017

MICCAI Workshop: Computational Diffusion MRI (CDMRI) 2017 (Quebec City, Canada, 10/09/2017). **Organisers:** Grussu F (UCL), Kaden E (UCL), Ning L (Harvard), Tax C (Cardiff University), Veraart J (NYU).

Data Harmonisation Challenge 2017

MICCAI Challenge: Diffusion MRI data harmonisation (part of CDMRI 2017). **Organisers:** Grussu F (UCL), Kaden E (UCL), Ning L (Harvard), Tax C (Cardiff University), Veraart J (NYU).

Spinal Cord Grey Matter Segmentation Challenge 2016

Challenge: Grey Matter Segmentation: What's there and What's next? (part of SC MRI Workshop 2016). **Organisers:** Cohen-Adad J, Prados F, Landman B, Wheeler-Kingshott C, Summers P, Dupont S, Yiannakas M, Smith S, Gergely D, DeLeener B, Grussu F.

SCIENTIFIC PEER REVIEWING

2016-present

Reviewer for Bipolar Disorders; Functional Neurology; Frontiers; Human Brain Mapping; Journal of Magnetic Resonance Imaging; Journal of Neuroscience Methods; Magnetic Resonance in Medicine; NeuroImage; Medical Image Analysis; IEEE Transactions on Medical Imaging; npj Precision Oncology; NMR in Biomedicine; Scientific Data; 2019, 2021, 2022 ISMRM annual meetings; 2021 ISMRM Iberian Chapter post-grad meeting; 2021, 2025 Iberian Chapter annual meeting; 2019, 2020 MICCAI workshop on Computational Diffusion MRI.

OFFICIAL STUDENT SUPERVISION — COMPLETED

PhD, VHIO and UB 2026b

Project co-supervision: *"Histology-informed diffusion MRI simulations and artificial intelligence for cancer microstructure characterization"*. **Degree:** PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). **Supervised in:** 01/10/2022-28/10/2026 (completed). **Final mark:** excellent; international PhD award.

MSc, VHIO and UAB 2026

Project co-supervision: "*Non-Invasive Tumour Microstructure Estimation Using Diffusion MRI: A Multi- Cohort Evaluation of Histo-μSim*". **Degree:** Master's in "Bioninformatics", Universitat Autònoma de Barcelona (UAB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). **Supervised in:** 01/02/2026-14/07/2026 (completed). **Final thesis mark:** 9.69/10.

PhD, VHIO and UB 2026a

Project co-supervision: *"A simulation framework for characterizing cancer microvasculature with histology-informed diffusion MRI"*. **Degree:** PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). **Supervised in:** 30/09/2022-23/04/2026 (completed). **Final mark:** excellent and *cum laude*; international PhD award.

Msc, VHIO and TU Delft 2023

Project co-supervision: *"Unraveling tumour microstructure through diffusion MRI using histology-powered artificial intelligence"*. **Degree:** Master's in "Biomedical Engineering", Delft University of Technology (TU Delft), (Delft, The Netherlands). Project carried out at VHIO, Barcelona (Spain). **Supervised in:** 09-2022/06-2023 (completed).

MSc, VHIO and UI La Rioja 2023

Project co-supervision: "*Prediction of advanced biomarkers from clinical diffusion Magnetic Resonance Imaging*". **Degree:** Master's in "Artificial Intelligence", Universidad Internacional de La Rioja (Logroño, Spain). Project carried out at VHIO, Barcelona (Spain). **Supervised in:** 2022/2023 (completed).

MRes, UCL 2019b

Project co-supervision: "*Improving the differential diagnosis between Neuromyelitis Optica Spectrum Disorder and Multiple Sclerosis using MRI*". **Degree:** MRes in "Clinical Neuroscience", Institute of Neurology, University College London (UCL, London, UK). **Supervised in:** 12/2018-06/2019 (completed).

MSc, UCL 2019a

Project co-supervision: "*Evaluation of quantitative MRI indices reproducibility*

across scanner upgrade". **Degree:** MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). **Supervised in:** 12/2018-06/2019 (completed).

MSc, UCL 2018

Project co-supervision: "*Investigation of multi-component T1 relaxation at 3 Tesla*". **Degree:** MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). **Supervised in:** 12/2017-06/2018 (completed).

MSc, UCL 2016

Project co-supervision: "*Evaluation of strategies for co-registration between quantitative and anatomical magnetic resonance images of the human spinal cord*". **Degree:** MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). **Supervised in:** 12/2015-06/2016 (completed).

OFFICIAL STUDENT SUPERVISION — ONGOING

PhD, VHIO and UB 2023-27

Project co-supervision: "*Novel Magnetic Resonance Imaging biomarkers for precision medicine in oncology: integrating multi-omics and real-world data strategies*". **Degree:** PhD programme in Biomedicine (research line in biomedical engineering), Universitat de Barcelona (UB; Barcelona, Spain). Project carried out at VHIO, Barcelona (Spain). **Supervised in:** 2023-2027 (ongoing).

TEACHING EXPERIENCE

UB Lecture

Lecture: "*Advanced medical imaging: a journey through engineering, physics and computer science*". **Subject/module:** "Artificial intelligence in biomedical engineering". **Degree:** BEng in "Biomedical engineering", Universitat de Barcelona (Barcelona, Spain). **Conveyed:** 12/05/2026. **Face-to-face teaching:** 2 hours.

UVi-UCC Lecture and Workshop

Lecture and Workshop: "*Medical image analysis*". **Subject/module:** "Interactomics". **Degree:** MSc in "Omics Data Analysis", Universitat de Vic-Universitat Central de Catalunya, UVic-UCC (Vic and Barcelona, Spain). **Conveyed:** 17/03/2026. **Face-to-face teaching:** 4 hours.

UCL Lecture

Lecture: "*Image optimisation: SNR, CNR and sources of artifacts*". **Degree:** MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). **Conveyed:** 2019/20, 2017/18, 2016/17, 2015/16. **Face-to-face teaching:** 4 hours (1 hour per year). **Additional tasks:** exam preparation and assessment.

UCL Workshop

Workshop: Hands-on with a portable MRI scanner. **Degree:** MSc in "Advanced neuroimaging", Institute of Neurology, UCL (London, UK). **Conveyed:** 2017/18, 2016/17, 2015/16. **Face-to-face teaching:** 12 hours (4 hour per year).

UniPV Workshop

Workshop: "*Model fitting for quantitative MRI*". **Degree:** MEng in "Biomedical Engineering", University of Pavia (Pavia, Italy). **Conveyed:** 2016/17. **Face-to-face teaching:** 2 hours. **Additional tasks:** exam preparation and assessment.

UCL Lecture

Lecture: "*Magnetic resonance image formation*". **Degree:** MSc in "Advanced biomedical imaging", Centre for Advanced Biomedical Imaging, UCL (London, UK). **Conveyed:** 2015/16. **Face-to-face teaching:** 1 hour.

OPEN SCIENCE

Code released through GitHub (http://fragrussu.github.io and http://github.com/radiomicsgroup)

GitHub 2024c

Histo-μSim: histology-informed cancer diffusion MRI (link).

GitHub 2024b

SpinFlowSim: diffusion MRI simulator in vascular networks (link).

GitHub 2024a

BodyMRITools: python code for body diffusion MRI processing (link).

GitHub 2022 **MChepato**: Code and synthetic data for Grussu et al, MRM 2022 (link). Available in Zenodo as record 6645258, doi: 10.5281/zenodo.6645258.

GitHub 2020b **qMRI-Net**: MRI signal model fitting based on artificial intelligence (link).

GitHub 2020a **SARDU-Net**: data-driven, model-free quantitative MRI protocol design (link).

GitHub 2019b **MyRelax**: tools for myelin and relaxation MRI analyses (link to the latest version). Version 1.0.0 available in Zenodo as record 4561898, doi: 10.5281/zenodo.4561898.

GitHub 2019a **MRITools**: tools for handling and managing research MRI scans (link).

GitHub 2016 **StructureTensorToolbox**: tools for analysis of 2D histological images (link).

Data sets released through Zenodo

Zenodo 2024b "*Histology-informed microstructural diffusion simulations for MRI cancer characterisation (Histo-µSim): ex vivo mouse data*". Grussu F, Grigoriou A, Macarro C and Perez-Lopez R. Data set. Zenodo record 14559355, 2024, doi: 10.5281/zenodo.14559355.

Zenodo 2024a "*Histology-informed microstructural diffusion simulations for MRI cancer characterisation (Histo-µSim): histology substrates*". Grigoriou A, Macarro C, Perez-Lopez R and Grussu F. Data set. Zenodo record 14559103, 2024, doi: 10.5281/zenodo.14559103.

DISSEMINATION TO THE WIDER PUBLIC

2024b Lab demonstration: "*Uso de la inteligencia artificial en la investigación contra el cáncer*" ("*Using artificial intelligence in cancer research*") at the 2024 European Researchers' Night (27/09/2024), CosmoCaixa, Barcelona (Spain).

2024a Video: "*Resonancia magnética para combatir el cáncer*" ("*Magnetic Resonance Imaging to fight cancer*"), released through the YouTube channel of the Vall d'Hebron Institute of Oncology (Barcelona, Spain) (link, video in Spanish).

2022b Science fair: "*Demostración de análisis avanzado de imágenes por resonancia magnética y datos co-localizados de microscopía*" ("*Demonstration of advanced analysis of MR images and co-localised microscopy*") at the 2022 European Researchers' Night (30/09/2022), CosmoCaixa, Barcelona (Spain).

2022a "*Innovating Magnetic Resonance Imaging to fight diseases*". Università della Terza Età (UniTre), Mòguru, Italy (12/04/2021). Talk given to a general audience in Sardinian language.

2017 "*Why to get vaccinated and avoid misinformation: the scientific method in modern medicine*". Event for a general audience in Italian and Sardinian, Mòguru, Italy (17/08/2017). Organisers: Grussu F, Tur C, Coccollone E, Broccia S.

2015 Participation at *MS Frontiers 2015* (29-30/06/2015), organised by the UK Multiple Sclerosis Society and bringing together researchers and Multiple Sclerosis patients.

2013 UCL stall at *Science Uncovered*, 2013 European Researcher's Night (28/09/2013), London Science Museum.

CONFERENCE & WORKSHOP PROCEEDINGS: FIRST AUTHORSHIP

ECR 2026 "*Response assessment and prognosis prediction in cancer immunotherapy using multi-parametric MRI: preliminary evidence*". Grussu F et al. European Conference of Radiology (ECR) 2026, C-16681 (EPOS^{TM}oral + digital poster presentation, 07/03/2026), doi: 10.26044/ecr2026/C-16681.

ESMRMB 2024 — "*Histological interpretation of Susceptibility-Perturbation MRI in human tumours of the liver*". Grussu F et al. Proc of the European Society for Magnetic Resonance in Medicine and Biology (ESMRMB) 2024 annual meeting (traditional poster presentation).

ISMRM 2024b — "*Histology-informed biophysical diffusion MRI model selection for enhanced liver cancer immunotherapy assessment*". Grussu F et al. Proc of the International Society for Magnetic Resonance in Medicine (ISMRM) 2024 annual meeting (**oral presentation**).

ISMRM 2024a — "*Two-axon population (TAP) modelling for large axon diffusion imaging in the peripheral nervous system*". Grussu F et al. Proc of ISMRM 2024 (**oral presentation**).

ISMRM Iberian 2023 — "*Extra-cellular liver diffusion modelling at high b-value: a preclinical MRI-histology study*". Grussu F et al. Iberian Chapter of the ISMRM 2023 (**oral presentation**).

ISMRM 2022b — "*Histological correlates of DR-HIGADOS microstructural metrics in the mouse and human liver*". Grussu F et al. Proc of ISMRM 2022 (power-pitch presentation).

ISMRM 2022a — "*Inter-scanner reproducibility and variability assessment of advanced liver diffusion MRI metrics*". Grussu F et al. Proc of ISMRM 2022 (d-poster presentation).

ISMRM 2021b — "*DR-HIGADOS: a new diffusion-relaxation framework for clinically feasible microstructural imaging of the liver*". Grussu F et al. Proc of ISMRM 2021 (**oral presentation**, ***Magna cum Laude*** award).

ISMRM 2021a — "*Investigating the relationship between diffusion MRI signal cumulants and hepatocyte microstructure at fixed diffusion time*". Grussu F et al. Proc of ISMRM 2021 (**d-poster presentation**).

ISMRM 2020 — "*SARDU-Net: a new method for model-free, data-driven experiment design in quantitative MRI*". Grussu F et al. Proc of ISMRM 2020 (**power-pitch presentation**, ***Magna cum Laude*** award).

ISMRM 2019 — "*Clinically viable g-ratio imaging with unified readout at 3T: evaluation and comparison*". Grussu F et al. Proc of ISMRM 2019 (e-poster presentation).

ISMRM 2018b — "*Magnitude versus complex-valued images for spinal cord diffusion MRI: which one is best?*". Grussu F et al. Proc of ISMRM 2018 (**oral presentation**, ***Magna cum Laude*** award).

ISMRM 2018a — "*A unified signal readout improves denoising of multi-modal spinal cord MRI*". Grussu F et al. Proc of ISMRM 2018 (poster presentation).

ISMRM 2017b — "*Origin of the time dependence of the diffusion-weighted signal in spinal cord white matter*". Grussu F et al. Proc of ISMRM 2017 (**oral presentation**).

ISMRM 2017a — "*A unified signal readout for reproducible multimodal characterisation of brain microstructure*". Grussu F et al. Proc of ISMRM 2017 (e-poster presentation, ***Magna cum Laude*** award, finalist at the Diffusion Study Group competition).

Brain School 2017 — "*Whole-brain macromolecular tissue volume mapping: A comparison of imaging readouts at 3 Tesla*". Grussu F et al. School of Brain Cells and Circuits "Camillo Golgi". published in Frontiers, ISBN 978-288945-584-3 (**poster presentation**).

ISMRM Scientific Workshop 2016 — "*Optimal histological quantification of neurite orientation dispersion for the validation of diffusion MRI*". Proc of ISMRM Scientific workshop "Breaking the barriers of diffusion MRI" (poster presentation + Power Pitch).

ISMRM 2016 — "*Axon diameter distribution influences diffusion-derived axonal density estimation in the human spinal cord:* in silico *and* in vivo *evidence*". Grussu F et al. ISMRM 2016 (poster presentation, finalist at the Diffusion Study Group competition).

ECTRIMS 2015 — "*Quantitative histological validation of NODDI MRI indices of neurite morphology in*

multiple sclerosis spinal cord". Grussu F et al. European Committee for Research and Treatment of Multiple Sclerosis (ECTRIMS) 2015 (poster presentation, short-listed for poster prize competition).

MS Frontiers 2015 "*Histological correlates of NODDI in the multiple sclerosis spinal cord*". Grussu F et al. MS Frontiers 2015, annual scientific meeting of the UK Multiple Sclerosis Society (**oral and poster presentation**).

ISMRM 2015b "*Quantitative histological correlates of NODDI orientation dispersion estimates in the human spinal cord*". Grussu F et al. Proc of ISMRM 2015 (**oral presentation**, ***Magna cum Laude*** award).

ISMRM 2015a "*Histological metrics confirm microstructural characteristics of NODDI indices in multiple sclerosis spinal cord*". Grussu F et al. Proc of ISMRM 2015 (**oral presentation**, ***Magna cum Laude*** award).

British Chapter of the ISMRM 2014 "*Characterisation of single-shell NODDI fitting in spinal cord grey and white matter*". Grussu F et al. Proc of the British Chapter of the ISMRM 2014 (poster presentation).

ISMRM 2014b "*Neurite orientation dispersion and density imaging of the cervical cord* in vivo". Grussu F et al. Proc of ISMRM 2014 (poster presentation).

ISMRM 2014a "*Single-shell diffusion MRI NODDI with* in vivo *cervical cord data*". Grussu F et al. Proc of ISMRM 2014 (poster presentation).

ISMRM Workshop 2013 "In vivo *estimation of neuronal orientation dispersion and density of the human spinal cord*". ISMRM Scientific workshop "Multiple sclerosis as a whole-brain disease" (**oral presentation**).

ISMRM 2013 "*Towards spinal cord microstructure mapping with the neurite orientation dispersion and density imaging*". Grussu F et al. Proc of ISMRM 2013 (poster presentation, finalist at the White Matter Study Group poster competition).

CONFERENCE & WORKSHOP PROCEEDINGS: SENIOR AUTHORSHIP

ESMRMB 2026c "*Towards the Acquisition-Independence of Microvascular Biomarkers in Diffusion MRI: a Feasibility Study at 0.55T*". Voronova A, ..., and Grussu F. Proc of ESMRMB 2026, PG.226 (poster presentation).

ESMRMB 2026b "*Non-invasive glioma grading through biophysical diffusion MRI modelling:the complementary role of Histo-µSim and SANDI*". Monetta A, ..., and Grussu F. Proc of ESMRMB 2026, PG.400 (poster presentation).

ESMRMB 2026a "*µNN: a physics-informed histology-MR neural network for microstructural diffusion imaging in cancer*". Grigoriou A, ..., and Grussu F. Proc of ESMRMB 2026, PG.031 (**oral presentation**).

ISMRM Iberian Chapter 2025 "*Simplicity is a virtue: histology-informed model comparison selects simple diffusion representations in a colorectal cancer metastasis specimen*". Grigoriou A, ..., and Grussu F. 2025 ISMRM Iberian Chapter annual meeting, p.49 (poster presentation). **2nd prize for the best poster presentation.**

ISMRM Workshop 2025b "*Which Microvascular Properties Can We Probe in Clinical Settings with Diffusion MRI?*". Voronova A, ..., and Grussu F. 2025 ISMRM Workshop on *40 Years of Diffusion: Past, Present and Future Perspectives* (**oral presentation**).

ISMRM Workshop 2025b "*Histology-Informed Microstructural Diffusion Simulations (Histo-µSim) for Enhanced Diffusion MRI Parameter Estimation in Cancer*". Grigoriou A, ..., and Grussu F. 2025 ISMRM Workshop on *40 Years of Diffusion: Past, Present and Future Perspectives* (power-pitch presentation), p. 202.

ESMRMB 2024b "*Biologically-realistic blood flow simulations reveal complex features of vascular IVIM*

signals". Voronova A, ..., and Grussu F. Proc of ESMRMB 2024 (traditional poster presentation).

ESMRMB 2024a "*Histology-informed cell size distribution mapping with diffusion MRI*". Grigoriou A, ..., and Grussu F. Proc of ESMRMB 2024 (traditional poster presentation).

ISMRM 2024b "*FlowSim: a blood flow simulator for histology-informed diffusion MRI micro-vasculature mapping in cancer*". Voronova A, ..., and Grussu F. Proc of ISMRM 2024 (power-pitch presentation, ***Summa cum Laude*** award, **Diffusion Study Group prize, best application (poster category)**).

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